US 20140273309A1

(19) **United States**

(12) **Patent Application Publication**

Niyogi et al.

(10) **Pub. No.: US 2014/0273309 A1**

(43) **Pub. Date: Sep. 18, 2014**

(54) **CONTROLLING RADICAL LIFETIMES IN A REMOTE PLASMA CHAMBER**

(71) Applicant: **Intermolecular, Inc.**, San Jose, CA (US)

(72) Inventors: **Sandip Niyogi**, San Jose, CA (US); **Sean Barstow**, San Jose, CA (US); **Jay Dedontney**, Prunedale, CA (US); **Chi-I Lang**, Cupertino, CA (US); **Ratsamee Limdulpaiboon**, San Jose, CA (US); **Martin Romero**, San Jose, CA (US); **Sunil Shanker**, Santa Clara, CA (US); **James Tsung**, Milpitas, CA (US); **J. Watanabe**, San Jose, CA (US)

(73) Assignee: **Intermolecular, Inc.**, San Jose, CA (US)

(21) Appl. No.: **14/051,287**

(22) Filed: **Oct. 10, 2013**

Related U.S. Application Data

(60) Provisional application No. 61/780,128, filed on Mar. 13, 2013.

Publication Classification

(51) **Int. Cl.**

H01L 21/66 (2006.01)

H01L 21/02 (2006.01)

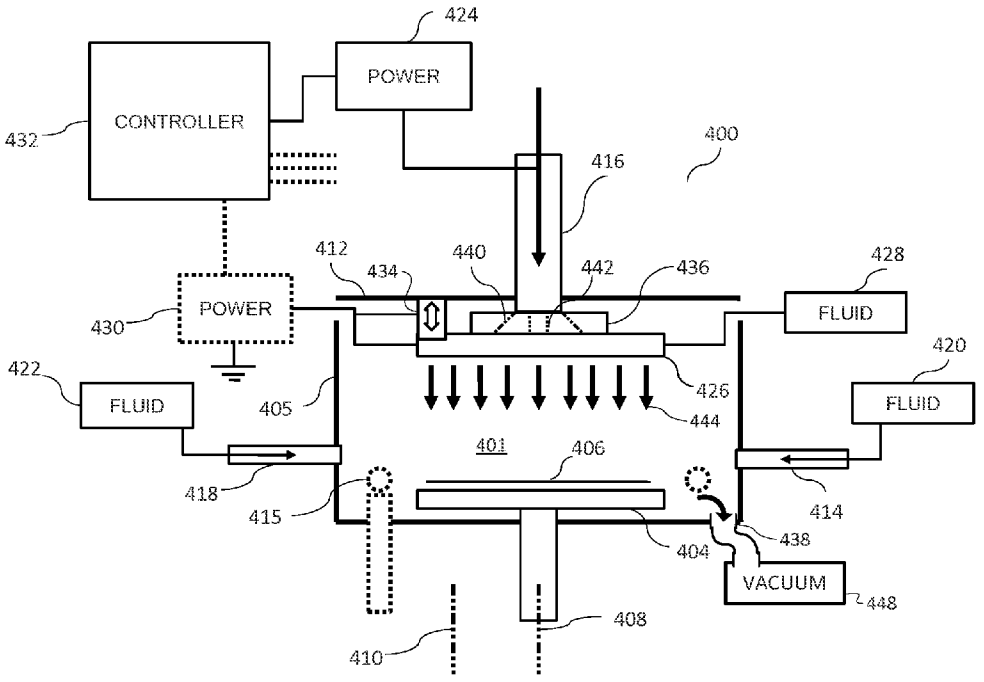
(52) **U.S. Cl.**

CPC ***H01L 22/10*** (2013.01); ***H01L 22/14*** (2013.01); ***H01L 22/12*** (2013.01); ***H01L 21/02104*** (2013.01)

USPC **438/16**; 438/17; 438/14; 134/1.2

(57) **ABSTRACT**

Remote-plasma treatments of surfaces, for example in semiconductor manufacture, can be improved by preferentially exposing the surface to only a selected subset of the plasma species generated by the plasma source. The probability that a selected species reaches the surface, or that an unselected species is quenched or otherwise converted or diverted before reaching the surface, can be manipulated by introducing additional gases with selected properties either at the plasma source or in the process chamber, varying chamber pressure or flow rate to increase or decrease collisions, or changing the dimensions or geometry of the injection ports, conduits and other passages traversed by the species. Some example processes treat surfaces preferentially with relatively low-energy radicals, vary the concentration of radicals at the surface in real time, or clean and passivate in the same unit process.



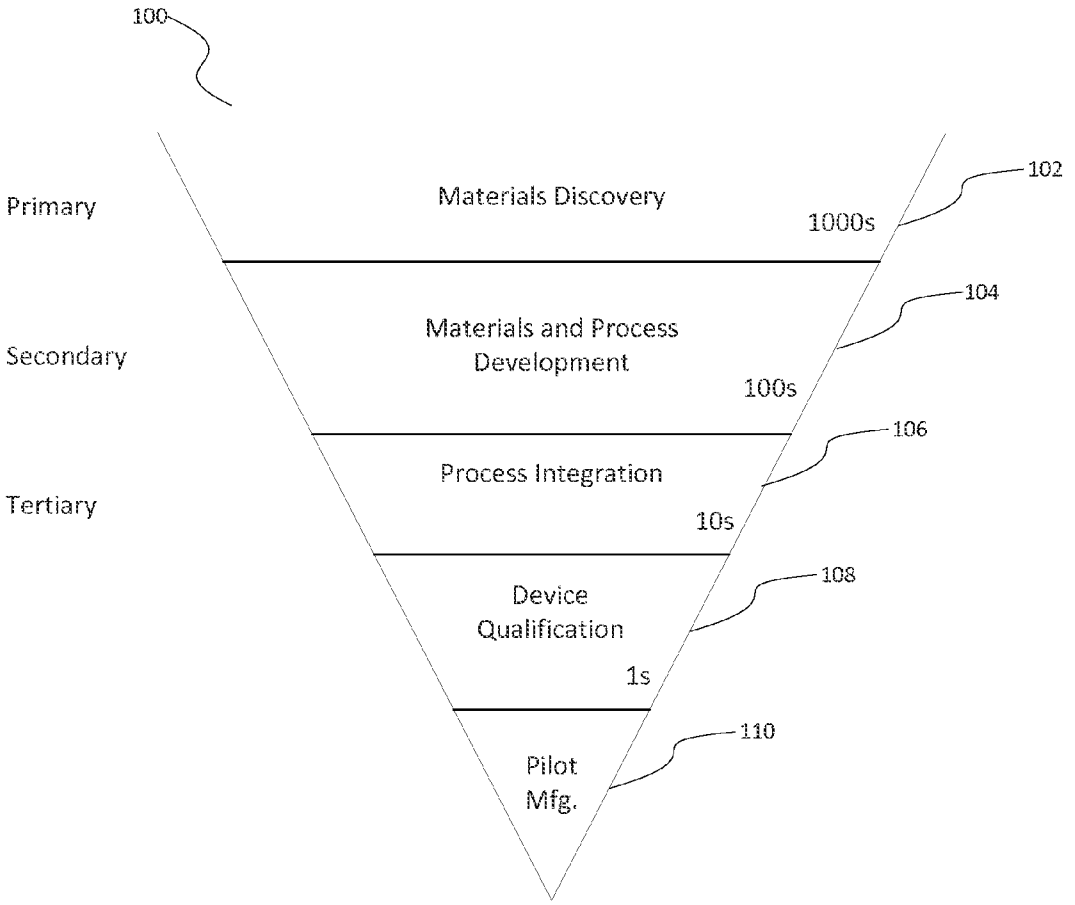

100
Primary
Secondary
Tertiary
Materials Discovery
1000s
102
Materials and Process Development
100s
104
Process Integration
10s
106
Device Qualification
1s
108
Pilot Mfg.
110

FIG. 1

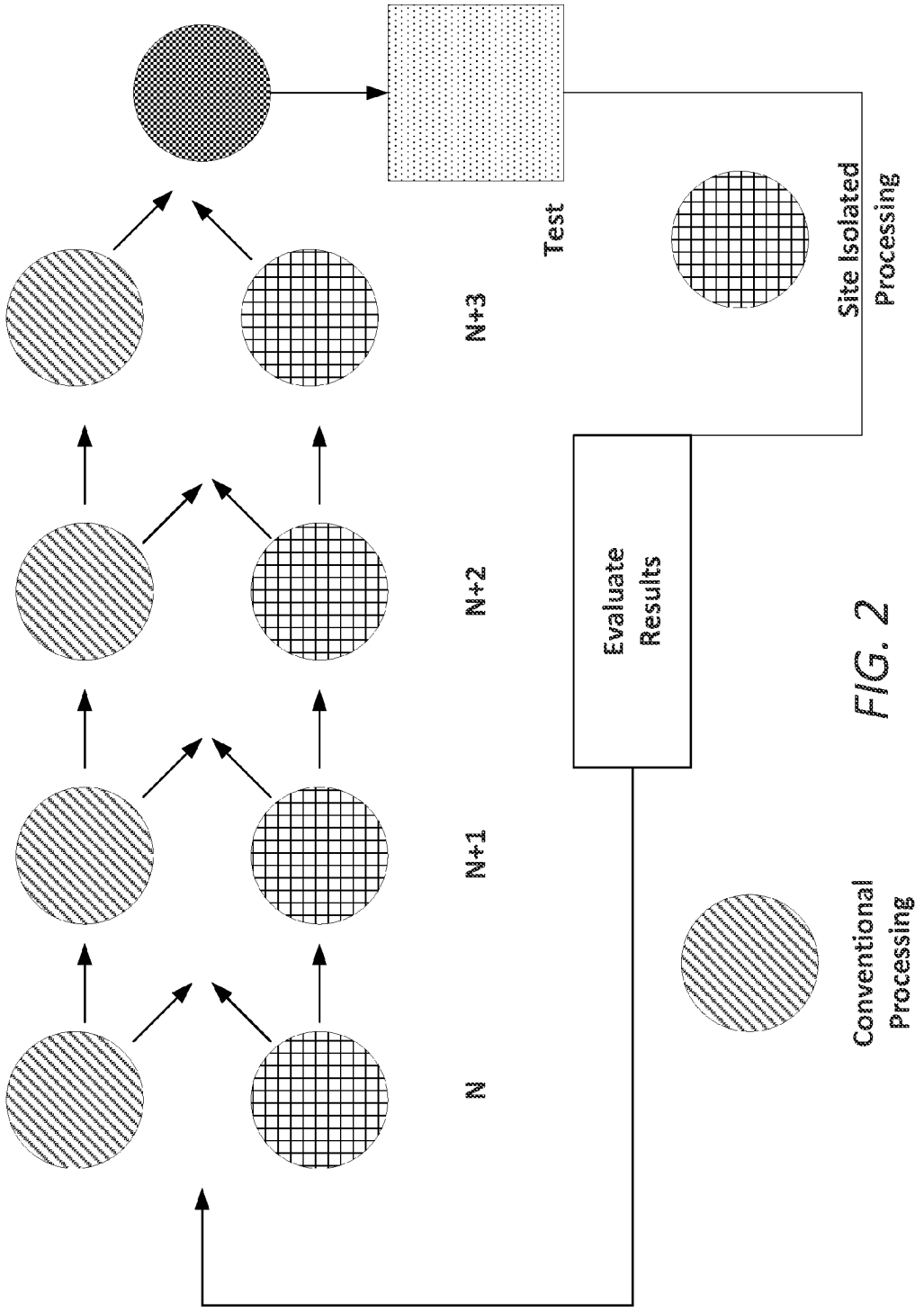

N
N+1
N+2
N+3
Test
Site Isolated Processing
Conventional Processing
Evaluate Results

FIG. 2

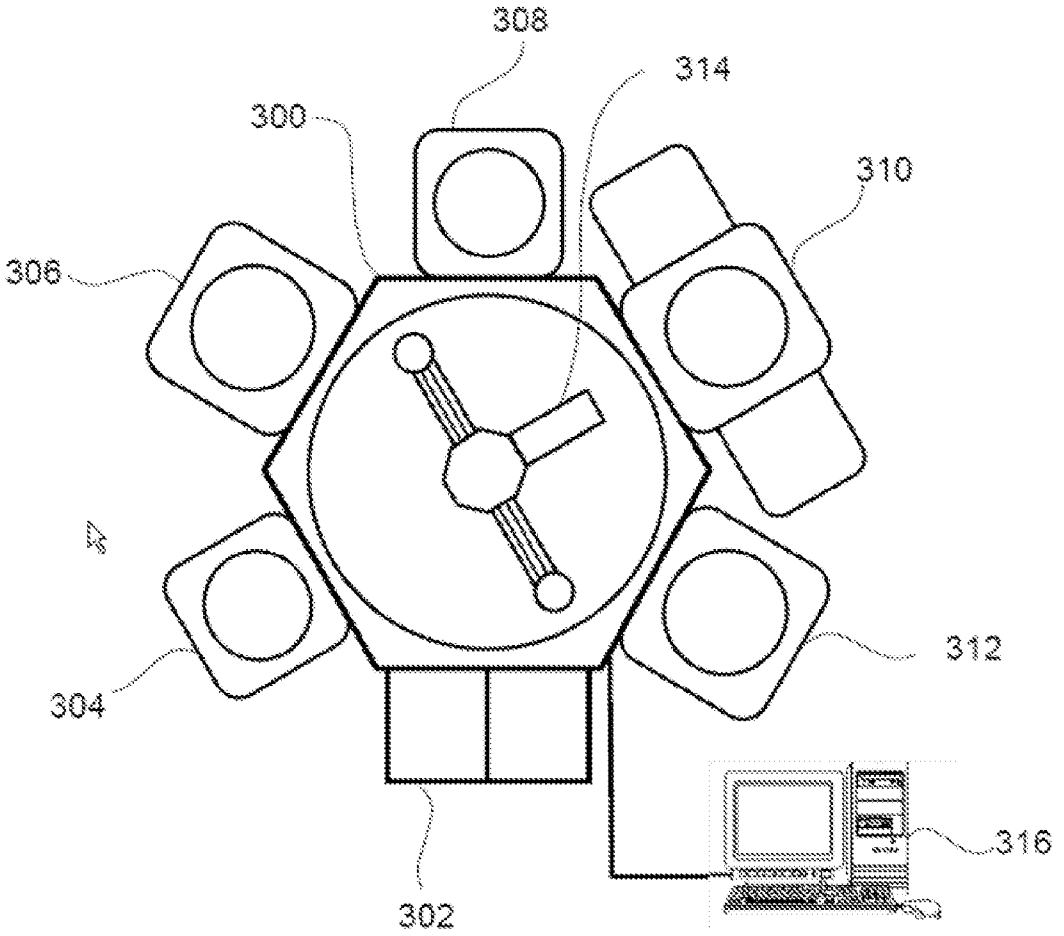


FIG. 3

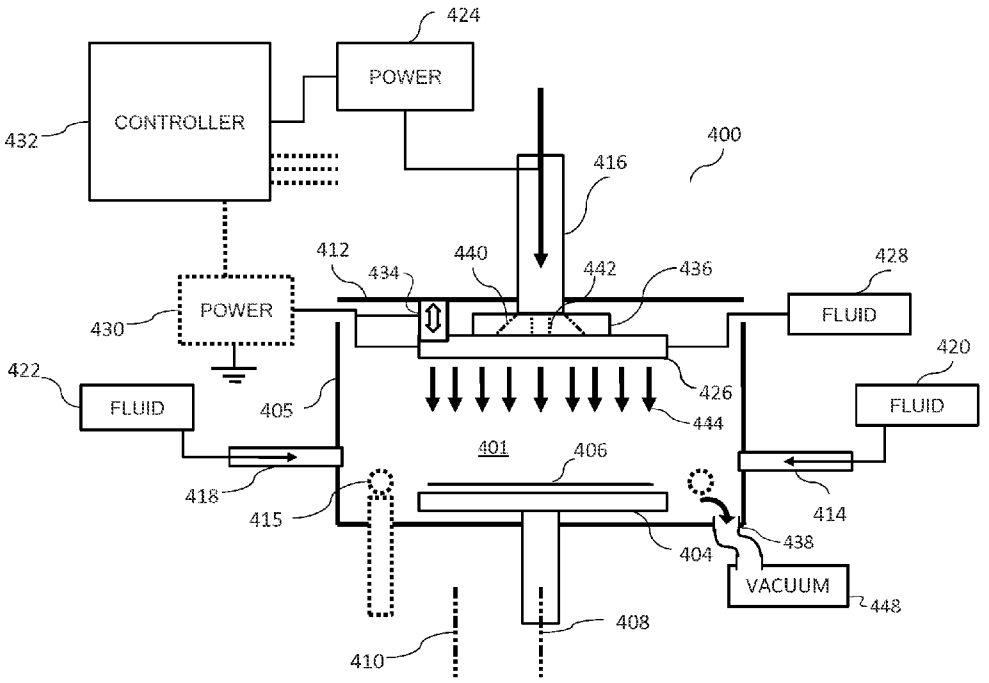

424
POWER
CONTROLLER
432
400
416
412
434
440
442
436
428
FLUID
POWER
430
420
422
FLUID
405
426
444
FLUID
401
406
418
415
414
438
404
VACUUM
448
408
410

FIG. 4

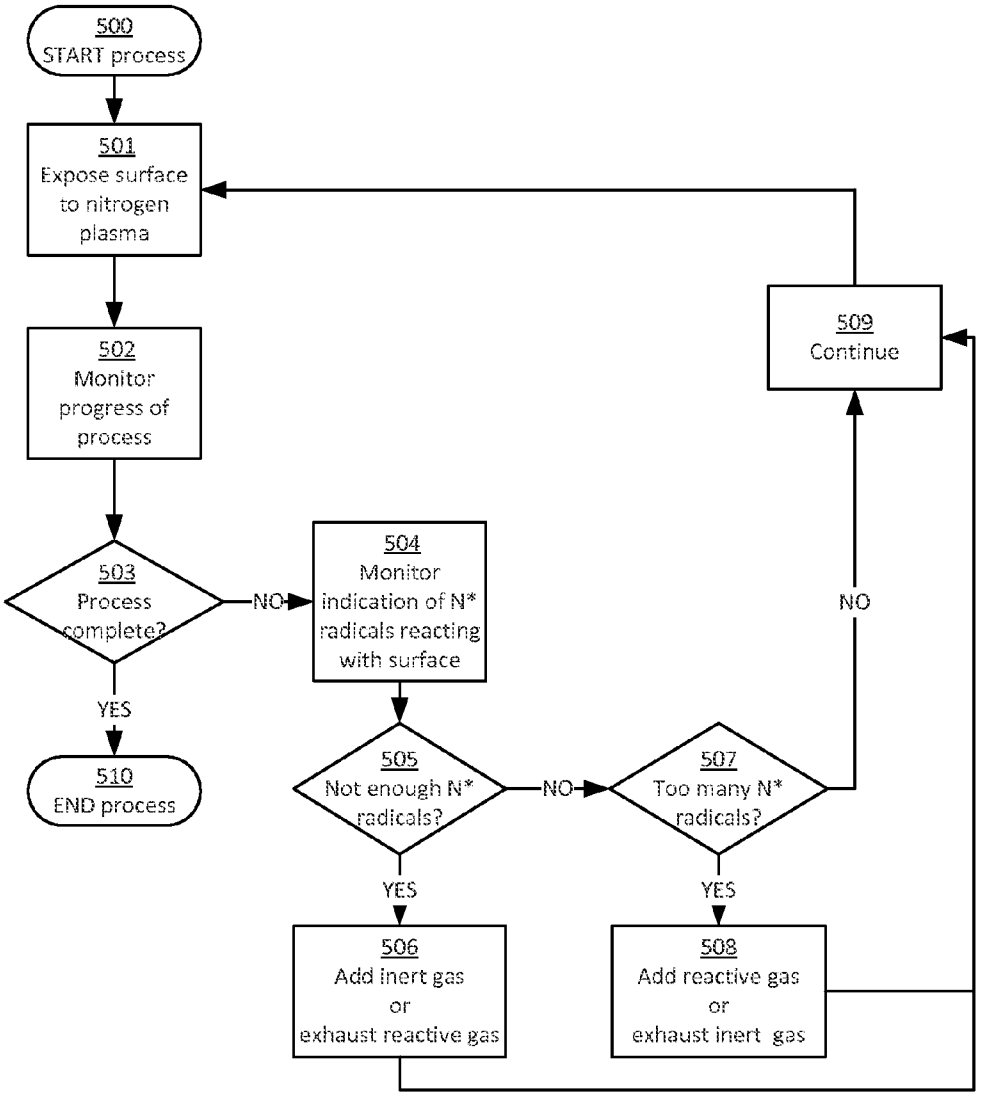

500
START process
501
Expose surface to nitrogen plasma
502
Monitor progress of process
503
Process complete?
NO
YES
504
Monitor indication of N* radicals reacting with surface
505
Not enough N* radicals?
NO
YES
506
Add inert gas or exhaust reactive gas
507
Too many N* radicals?
NO
YES
508
Add reactive gas or exhaust inert gas
509
Continue
510
END process

FIG. 5A

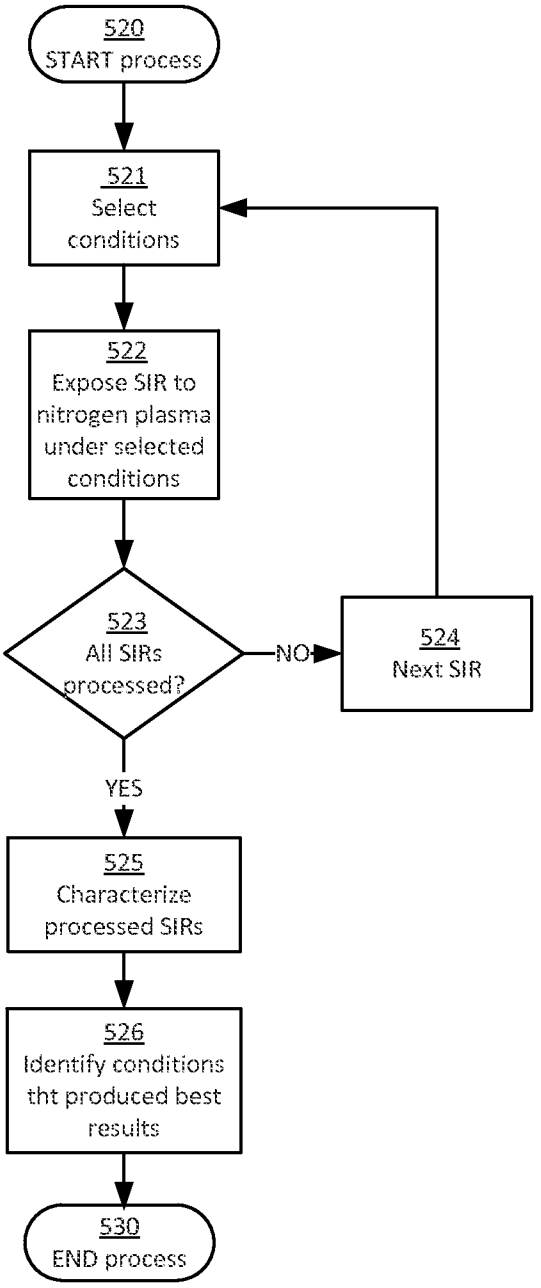

520
START process
521
Select conditions
522
Expose SIR to nitrogen plasma under selected conditions
523
All SIRs processed?
NO
524
Next SIR
YES
525
Characterize processed SIRs
526
Identify conditions tht produced best results
530
END process

FIG. 5B

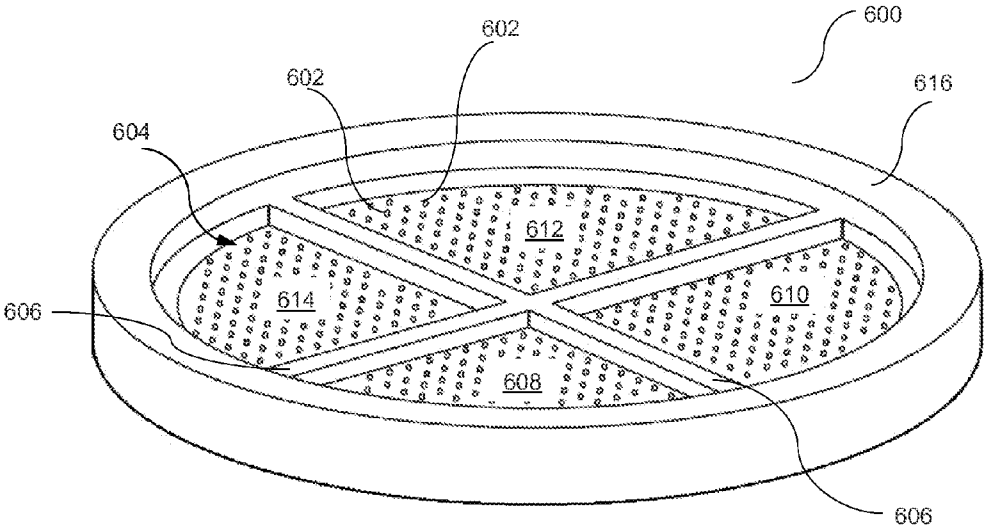


FIG. 6A

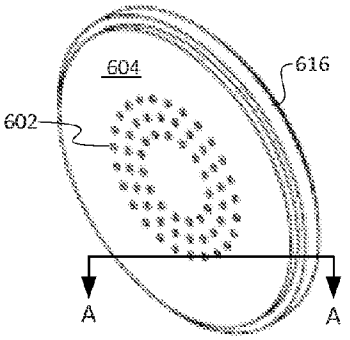


FIG. 6B

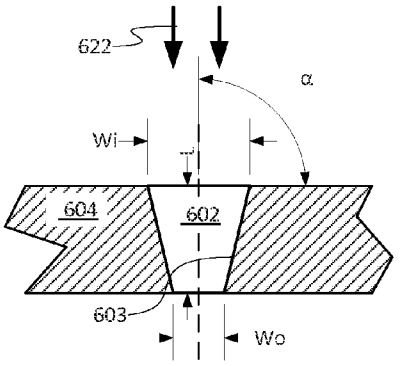


FIG. 6C

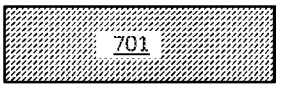


FIG. 7A

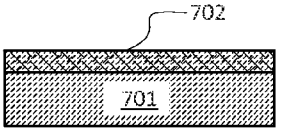


FIG. 7B

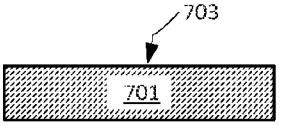


FIG. 7C

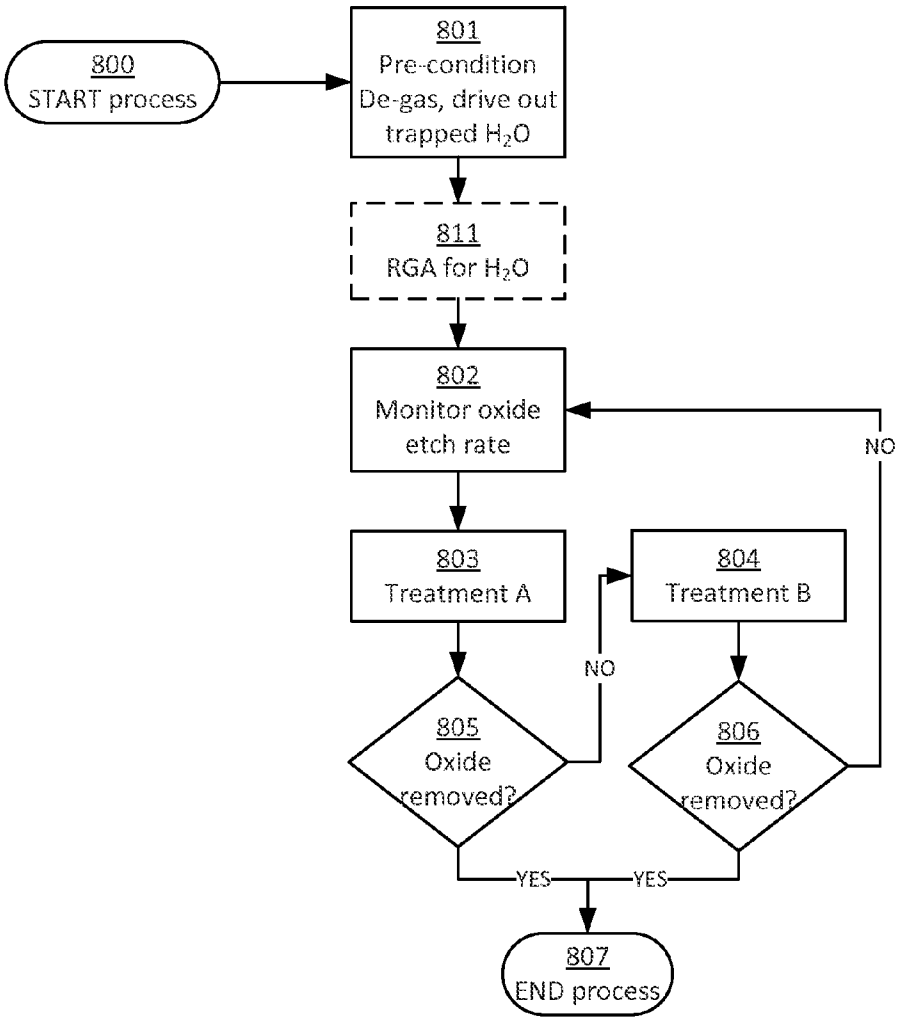


FIG. 8

CONTROLLING RADICAL LIFETIMES IN A REMOTE PLASMA CHAMBER

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to U.S. Provisional Pat. App. Ser. No. 61/780,128, filed 13 Mar. 2013 and incorporated herein by reference for all purposes.

BACKGROUND

[0002] Related fields include remote-plasma processing of substrates and thin films, and surface treatments in semiconductor manufacture.

[0003] The continued demand for smaller and smaller semiconductor devices has increased the importance of surface treatments. Reducing the size of a thin-film-based feature often increases its sensitivity to defects, roughness, contaminants, and other imperfections or inconsistencies in the interfaces between layers. Surface treatments before forming an additional layer on the surface can sometimes produce more desirable properties in the interface between the layers.

[0004] Plasma-processing techniques modify the properties of a surface (or of that portion of a film or bulk material that is very close to the surface) by exposing the surface to a plasma. Plasmas may be generated for industrial use by DC, RF, or microwave power sources in high vacuum, moderate vacuum, or near-atmospheric pressure. Plasmas may include a variety of plasma activated species; besides the ions, free electrons, and other charge-carriers that make the plasma electrically conductive, neutral particles and radicals may also be present.

[0005] Some plasma processes used in semiconductor manufacture (e.g. plasma-enhanced atomic layer deposition (PEALD), nitridation, plasma-doping, passivation, native-oxide removal) might produce improved performance, reliability, or consistency if the surface could be selectively exposed to only a subset of the plasma activated species generated by the plasma. In some cases, some of the species may counteract the intended effects of other species. In other cases, some species produce a desired effect on selected features fabricated on a substrate while other species can damage other nearby structures.

[0006] The ability to dynamically change the density of a given species reaching a surface during treatment is also desirable. For example, a plasma may contain both etchant and passivation species, and a surface being processed may be a "native" oxide film on a semiconductor. The etchant film may need to be strong to remove the oxide, but its full strength may damage the semiconductor underneath. The semiconductor surface, once exposed, may have dangling bonds that become voids or other defects, or promote the formation of another native oxide. If the etchant species could be reduced and the passivation species increased as the last of the native oxide is removed, a smoother oxide-free semiconductor surface might result.

[0007] Therefore, the semiconductor industry would benefit from the ability to select the dominant species impinging on the work surface, to adjust them in real time, and to monitor properties of both the plasma and the work surface. Removal of native oxides, a very common challenge, is one example of a process that could be improved by these abilities.

SUMMARY

[0008] The following summary presents some concepts in a simplified form as an introduction to the detailed description that follows. It does not necessarily identify key or critical elements and is not intended to reflect a scope of invention.

[0009] Plasma generation sources generate a variety of plasma activated species; for example, ions, electrons, and radicals. Some plasma treatments benefit from preferentially exposing the surface under treatment to a subset of the plasma activated species generated by the source. Some processes are primarily executed by a subset of the generated species, and the rest are superfluous. Some surfaces are damaged by the excess energy dissipated by higher-energy species but can be acceptably treated by lower-energy species. Some surfaces are undesirably affected by ions (e.g., unwanted dipoles or dangling bonds may be formed), but desirably affected by radicals. A subset of species intended to react with the substrate can be preferentially selected by increasing the probability that a the selected species will react with the substrate, or by decreasing the probability that other, unselected species will react with the substrate.

[0010] The probability may be increased for a selected species that promotes the intended treatment. Adding an inert gas may increase the number of selected species generated by collisions. The gas conductance (as distinct from electrical conductance) of showerheads, injection ports, and other conduits through which the plasma species pass can be optimized to give the selected species a travel time less than its expected lifetime.

[0011] The probability may be decreased for an unselected species; i.e., one that does not promote the intended treatment, or that inhibits the intended treatment, or that may damage the surface being treated. Adding a reactive gas may quench the unselected species and shorten its expected lifetime below the expected travel time to the surface. Increasing pressure and temperature or manipulating gas conductance to increase the frequency of collisions and decrease the mean free path may exclude unselected species with expected lifetimes shorter than the travel time to the surface and shorter than the expected lifetime of a selected species.

[0012] Nitridation (conversion of a metal, semiconductor, compound or alloy to its corresponding nitride), nitrogen doping, or other treatments with N* radicals (herein, an asterisk after an element's chemical symbol denotes a radical species of that element) can be controlled dynamically as the treatment continues. To increase the concentration of radicals at the surface, an inert gas is added to create more radicals through collision. To decrease the concentration of radicals at the surface, a reactive gas is added to quench the radicals before they reach the surface. Transitions can be made more gradual by exhausting one of the gases before adding the other.

[0013] Germanium and III-V materials (compound semiconductor materials containing elements from Groups III and V of the periodic table, e.g., gallium arsenide, gallium nitride) can be cleaned, including the removal of native oxides, and passivated by alternating exposure to O* and H* radicals. Ion bombardment causes unacceptable damage to these materials, so this process may benefit from exposure to a subset of plasma species that includes radicals and excludes ions.

BRIEF DESCRIPTION OF DRAWINGS

[0014] FIG. **1** illustrates a schematic diagram **100** for implementing combinatorial processing and evaluation using primary, secondary, and tertiary screening.

[0015] FIG. **2** is a simplified schematic diagram illustrating a general methodology for combinatorial process sequence integration that includes site-isolated processing and/or conventional processing.

[0016] FIG. **3** is a simplified schematic diagram of an example of an integrated high productivity combinatorial (HPC) system.

[0017] FIG. **4** is a simplified schematic diagram illustrating a processing chamber, or substrate processing tool configured to perform combinatorial processing.

[0018] FIGS. **5**A and **5**B are flowcharts of example processes for controlling a concentration of N* radicals reacting with a surface of a substrate.

[0019] FIGS. **6**A-**6**C illustrate embodiments of showerheads and their injection ports.

[0020] FIGS. **7**A-**7**C conceptually illustrate the formation and removal of a native oxide.

[0021] FIG. **8** is a flowchart of an example process for native-oxide removal using O* and H* radicals as the selected species.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

[0022] A detailed description of one or more example embodiments is provided below. To avoid unnecessarily obscuring the description, some technical material known in the related fields is not described in detail. Semiconductor fabrication generally requires many other processes before and after those described; this description omits steps that are irrelevant to, or that may be performed independently of, the described processes.

[0023] Substrate surfaces may be treated before depositing, growing, or otherwise forming additional layers or features. Alternatively, an intended outer surface may be treated to confer desirable chemical or physical properties.

[0024] In the described treatments, a remote plasma source generates a variety of plasma activated species such as electrons, ions, and radicals. The surface to be treated is preferentially exposed to a selected subset of those species, rather than to all of the generated species. The probability of a particular species reaching the surface under treatment is manipulated (i.e., species are selected or excluded) by introducing gases into the chamber or by changing the physical parameters and spatial relationships of hardware components in the process chamber. Increasing the probability of a selected species reaching the surface, or decreasing the probability of an unselected species reaching the surface, can improve a quality of the surface or a quality of an interface with an additional film or feature subsequently formed adjacent to the surface.

[0025] Some treatments that benefit from species selection or exclusion relate to removal of native oxides from substrates or films such as germanium (Ge) and III-V materials. Impinging ions can easily damage these materials, but O* and H* radicals remove the oxide and passivate the surface without damage. A pre-conditioning step may be used before the plasma treatment to remove water from the oxide and from chamber hardware such as the process kit and showerhead.

[0026] Combinatorial processing may be used to produce and evaluate different materials, chemicals, processes, and techniques, or build structures to determine how materials interact with existing structures, across multiple site-isolated regions (SIRs) on each substrate. These variations may relate to temperature, exposure time, layer thickness, chemical composition, humidity, and other process variables with a chemical composition kept constant, or the chemical composition may be varied.

[0027] Techniques to evaluate the effects of the variations include monitoring the substrate by measuring the sheet resistance of an underlying film (e.g., a titanium (Ti) film) and monitoring the optical emission spectroscopy of the plasma source. One or more oxide peaks may be monitored in the Fourier-transform infrared (FTIR) spectrum near the substrate. A method of evaluating a cleaning process involves creating a diode by depositing a conductive film on the treated surface. The diode's current vs. voltage (I-V) curve may then be measured and compared to that of a diode made from a known clean sample of the same material. Some treatments, such as those used to reduce current leakage in dielectric films, may be evaluated by measuring the C-V (capacitance vs. voltage) curve of a capacitor formed on the substrate (e.g., by sandwiching a dielectric layer between two conductive layers, and optionally etching to produce multiple capacitors).

[0028] Semiconductor manufacturing typically includes a series of processing steps such as cleaning, surface preparation, deposition, patterning, etching, thermal annealing, and other related unit processing steps. The precise sequencing and integration of the unit processing steps enables the formation of functional devices meeting desired performance metrics such as efficiency, power production, and reliability. As part of the discovery, optimization and qualification of each unit process, it is desirable to be able to test i) different materials, ii) different processing conditions within each unit process module, iii) different sequencing and integration of processing modules within an integrated processing tool, iv) different sequencing of processing tools in executing different process sequence integration flows, and (v) combinations thereof. In particular, there is a need to be able to test multiple materials, processing conditions, sequences of processing conditions, process sequence integration flows, and combinations (collectively, "combinatorial process sequence integration") on a single substrate rather than using a separate substrate for each combination of materials, processes, sequences, and flows. This can greatly increase the speed and reduce the cost of discovery, implementation, optimization, and qualification of the material(s), process(es), and process integration sequence(s) required for manufacturing.

[0029] Systems and methods for High Productivity Combinatorial (HPC) processing are described in U.S. Pat. No. 7,544,574 filed on Feb. 10, 2006, U.S. Pat. No. 7,824,935 filed on Jul. 2, 2008, U.S. Pat. No. 7,871,928 filed on May 4, 2009, U.S. Pat. No. 7,902,063 filed on Feb. 10, 2006, and U.S. Pat. No. 7,947,531 filed on Aug. 28, 2009 which are all herein incorporated by reference. Systems and methods for HPC processing are further described in U.S. patent application Ser. No. 11/352,077 filed on Feb. 10, 2006, claiming priority from Oct. 15, 2005, U.S. patent application Ser. No. 11/419,174 filed on May 18, 2006, claiming priority from Oct. 15, 2005, U.S. patent application Ser. No. 11/674,132 filed on Feb. 12, 2007, claiming priority from Oct. 15, 2005, and U.S.

patent application Ser. No. 11/674,137 filed on Feb. 12, 2007, claiming priority from Oct. 15, 2005 which are all herein incorporated by reference.

[0030] HPC processing techniques have been successfully adapted to wet chemical processing such as etching and cleaning. HPC processing techniques have also been successfully adapted to deposition processes such as physical vapor deposition (PVD), atomic layer deposition (ALD), and chemical vapor deposition (CVD). A variety of test and measurement techniques to evaluate the performance of these and other processes have been incorporated into HPC methods and, in some cases, into HPC apparatus.

[0031] FIG. **1** illustrates a schematic diagram **100** for implementing combinatorial processing and evaluation using primary, secondary, and tertiary screening. The schematic diagram **100** illustrates that the relative number of combinatorial processes run with a group of substrates decreases as certain materials and/or processes are selected. Generally, combinatorial processing includes performing a large number of processes during a primary screen, selecting promising candidates from those processes, performing the selected processing during a secondary screen, selecting promising candidates from the secondary screen for a tertiary screen, and so on. In addition, feedback from later stages to earlier stages can be used to refine the success criteria and provide better screening results.

[0032] For example, thousands of materials are evaluated during a materials discovery stage **102**. Materials discovery stage **102** is also known as a primary screening stage performed using primary screening techniques. Primary screening techniques may include dividing substrates into coupons and depositing materials using varied processes. The materials are then evaluated, and promising candidates are advanced to the secondary screen, or materials and process development stage, **104**. Evaluation of the materials is performed using metrology tools such as electronic testers and imaging tools (i.e., microscopes).

[0033] The materials and process development stage **104** may evaluate hundreds of materials (i.e., a magnitude smaller than the primary stage) and may focus on the processes used to deposit or develop those materials. Promising materials and processes are again selected, and advanced to the tertiary screen or process integration stage **106**, where tens of materials and/or processes and combinations are evaluated. The tertiary screen or process integration stage **106** may focus on integrating the selected processes and materials with other processes and materials.

[0034] The most promising materials and processes from the tertiary screen are advanced to device qualification **108**. In device qualification, the materials and processes selected are evaluated for high volume manufacturing, which normally is conducted on full substrates with production tools, but need not be conducted in such a manner. The results are evaluated to determine the efficacy of the selected materials and processes. If successful, the use of the screened materials and processes can proceed to pilot manufacturing **110**.

[0035] The schematic diagram **100** is an example of various techniques that may be used to evaluate and select materials and processes for the development of new materials and processes. The descriptions of screening stages **102-110** are arbitrary; the stages may overlap, occur out of sequence, or be described or performed in other ways.

[0036] This application benefits from High Productivity Combinatorial (HPC) techniques described in U.S. patent application Ser. No. 11/674,137 filed on Feb. 12, 2007, and incorporated for reference in its entirety for all purposes. Portions of the '137 application have been reproduced below to enhance the understanding of the present invention. The embodiments described herein enable the application of combinatorial techniques to process sequence integration in order to arrive at a globally optimal sequence of semiconductor manufacturing operations by considering interaction effects between the unit manufacturing operations, the process conditions used to effect such unit manufacturing operations, hardware details used during the processing, and material characteristics of components utilized within the unit manufacturing operations. Rather than only considering a series of local optima, i.e., the best conditions and materials for each manufacturing unit operation considered in isolation, the embodiments described below consider interaction effects introduced by the multitude of processing operations that are performed and the order in which such multitude of processing operations are performed when fabricating a device. A global optimum sequence order is therefore derived and as part of this derivation, the unit processes, unit process parameters and materials used in the unit process operations of the optimum sequence order are also considered.

[0037] The embodiments described further analyze a portion or sub-set of the overall process sequence used to manufacture, for example, a semiconductor device. Once the subset of the process sequence is identified for analysis, combinatorial process sequence integration testing is performed to optimize the materials, unit processes, hardware details, and process sequence used to build that portion of the device or structure. During the processing of some embodiments described herein, structures are formed on the processed substrate that are equivalent to the structures formed during actual production of the semiconductor device. For example, such structures may include, but would not be limited to, contact layers, buffer layers, absorber layers, or any other series of layers or unit processes that create an intermediate structure found on semiconductor devices.

[0038] While the combinatorial processing varies certain materials, unit processes, hardware details, or process sequences, the composition or thickness of the layers or structures or the action of the unit process, such as cleaning, surface preparation, deposition, surface treatment, etc. is substantially uniform through each discrete region. Furthermore, while different materials or unit processes may be used for corresponding layers or steps in the formation of a structure in different regions of the substrate during the combinatorial processing, the application of each layer or use of a given unit process is substantially consistent or uniform throughout the different regions in which it is intentionally applied. Thus, the processing is uniform within a region (inter-region uniformity) and between regions (intra-region uniformity), as desired. It should be noted that the process can be varied between regions, for example, where a thickness of a layer is varied or a material may be varied between the regions, etc., as desired by the design of the experiment.

[0039] The result is a series of regions on the substrate that contain structures or unit process sequences that have been uniformly applied within that region and, as applicable, across different regions. This process uniformity allows comparison of the properties within and across the different regions such that the variations in test results are due to the varied parameter (e.g., materials, unit processes, unit process parameters, hardware details, or process sequences) and not

the lack of process uniformity. In the embodiments described herein, the positions of the discrete regions on the substrate can be defined as needed, but are preferably systematized for ease of tooling and design of experimentation. In addition, the number, variants and location of structures within each region are designed to enable valid statistical analysis of the test results within each region and across regions to be performed.

[0040] FIG. **2** is a simplified schematic diagram illustrating a general methodology for combinatorial process sequence integration that includes site-isolated processing and/or conventional processing. In some embodiments, the substrate is initially processed using conventional process N. In some exemplary embodiments, the substrate is then processed using site-isolated process N+1. During site-isolated processing, an HPC module may be used, such as the HPC module described in U.S. patent application Ser. No. 11/352,077 filed on Feb. 10, 2006. The substrate can then be processed using site-isolated process N+2, and thereafter processed using conventional process N+3. Testing is performed and the results are evaluated. The testing can include physical, chemical, acoustic, magnetic, electrical, optical, etc. tests. From this evaluation, a particular process from the various site-isolated processes (e.g. from steps N+1 and N+2) may be selected and fixed so that additional combinatorial process sequence integration may be performed using site-isolated processing for either process N or N+3. For example, a next process sequence can include processing the substrate using site-isolated process N, conventional processing for processes N+1, N+2, and N+3, with testing performed thereafter.

[0041] Various other combinations of conventional and combinatorial processes can be included in the processing sequence of FIG. **2**. The combinatorial process sequence integration can be applied to any desired segments and/or portions of an overall process flow. Characterization, including physical, chemical, acoustic, magnetic, electrical, optical, etc. testing, can be performed after each process operation, and/or series of process operations within the process flow as desired. The feedback provided by the testing is used to select certain materials, processes, process conditions, and process sequences and eliminate others. Furthermore, the above flows can be applied to entire monolithic substrates, or portions of monolithic substrates such as coupons.

[0042] Under combinatorial processing operations, the processing conditions in different regions can be controlled independently. Consequently, process material amounts, reactant species, processing temperatures, processing times, processing pressures, processing flow rates, processing powers, processing reagent compositions, the rates at which the reactions are quenched, deposition order of process materials, process sequence steps, hardware details, etc., can be varied from region to region on the substrate.

[0043] Thus, for example, when exploring materials, a processing material delivered to a first and second region can be the same or different. If the processing material delivered to the first region is the same as the processing material delivered to the second region, this processing material can be offered to the first and second regions on the substrate at different concentrations. In addition, the material can be deposited under different processing parameters. Parameters which can be varied include, but are not limited to, process material amounts, reactant species, processing temperatures, processing times, processing pressures, processing flow rates, processing powers, processing reagent compositions, the rates at which the reactions are quenched, atmospheres in which the processes are conducted, an order in which materials are deposited, hardware details of the gas distribution assembly, etc. It should be appreciated that these process parameters are exemplary and not meant to be an exhaustive list as other process parameters commonly used in semiconductor manufacturing may be varied.

[0044] As mentioned above, within a region, the process conditions are substantially uniform, in contrast to gradient processing techniques which rely on the inherent non-uniformity of the material deposition. That is, the embodiments described herein locally perform the processing in a conventional manner, e.g., substantially consistent and substantially uniform, while globally over the substrate, the materials, processes, and process sequences may vary. Thus, the testing will find optima without interference from process variation differences between processes that are meant to be the same. It should be appreciated that a region may be adjacent to another region in some embodiments or the regions may be isolated and, therefore, non-overlapping. When the regions are adjacent, there may be a slight overlap wherein the materials or precise process interactions are not known; however, a portion of the regions, normally at least 50% or more of the area, is uniform and all testing occurs within that region. Further, the potential overlap is only allowed with materials or processes that will not adversely affect the result of the tests. Both types of regions are referred to herein as regions or discrete regions.

[0045] FIG. **3** is a simplified schematic diagram of an example of an integrated high productivity combinatorial (HPC) system. HPC system includes a frame **300** supporting a plurality of processing modules. It should be appreciated that the frame **300** may be a unitary frame in accordance with some embodiments. In some embodiments, the environment within the frame **300** is controlled. Load lock/factory interface **302** provides access into the plurality of modules of the HPC system. Robot **314** provides for the movement of substrates (and masks) between the modules and for the movement into and out of the load lock **302**. Modules **304-312** may be any set of modules and preferably include one or more combinatorial modules. For example, module **304** may be an orientation/degassing module, module **306** may be a clean module, either plasma or non-plasma based, modules **308** and/or **310** may be combinatorial/conventional dual purpose modules. Module **312** may provide conventional clean or degas as necessary for the experiment design.

[0046] Any type of chamber or combination of chambers may be implemented and the description herein is merely illustrative of one possible combination and not meant to limit the potential chamber or processes that can be supported to combine combinatorial processing or combinatorial plus conventional processing of a substrate or wafer. In some embodiments, a centralized controller, i.e., computing device **316**, may control the processes of the HPC system, including the power supplies and synchronization of the duty cycles described in more detail below. Further details of one possible HPC system are described in US application Ser. Nos. 11/672,478 and 11/672,473. With HPC system, a plurality of methods may be employed to deposit material upon a substrate employing combinatorial processes.

[0047] FIG. **4** is a simplified schematic diagram illustrating a processing chamber, or substrate processing tool configured to perform combinatorial processing. The processing chamber **400** is defined by a housing that includes a sidewall **405** and a lid **412** enclosing a chamber interior **401**. Processing

chamber **400** also includes a substrate support **404** configured to hold a substrate **406**. The substrate support **404** may be any known substrate support, including but not limited to a vacuum chuck, electrostatic chuck or other known mechanisms. The substrate support **404** is capable of both rotating around its own central axis **408** (referred to as "rotation" axis, which is congruent with a central axis of the substrate **406**), and rotating around a second axis **410** (referred to as "revolution" axis). Other substrate supports, such as an XY table, can also be used for site-isolated processing. In addition, the substrate support **404** may move in a vertical direction, i.e., away from or towards lid **412**. Rotation and movement in the vertical direction may be achieved through known drive mechanisms which include magnetic drives, linear drives, worm screws, lead screws, a differentially pumped rotary feed through drive, etc. A power source **424** provides power to plasma generation source **416**. It should be appreciated that power source **424** may output a direct current (DC) power supply, a pulsed DC power supply, or a radio frequency (RF) power supply.

[0048] The substrate **406** may be a conventional round 200 mm, 300 mm substrate, or any other larger or smaller substrate/wafer size. In some embodiments, the substrate **406** may be a square, rectangular, or other shaped substrate. One skilled in the art will appreciate that the substrate **406** may be a blanket substrate, a coupon (e.g., partial wafer), or even a patterned substrate having predefined regions. In some embodiments, the substrate **406** may have regions defined through the processing described herein. The term "region" is used herein to refer to a localized area on a substrate which is, was, or is intended to be used for processing or formation of a selected material. The region can include one region and/or a series of regular or periodic regions predefined on the substrate. The region may have any convenient shape, e.g., circular, rectangular, elliptical, wedge-shaped, etc. In the semiconductor field a region may be, for example, a test structure, single die, multiple dies, portion of a die, other defined portion of substrate, or an undefined area of a substrate, e.g., blanket substrate which is defined through the processing.

[0049] The chamber **400** in FIG. **4** includes a lid **412**, through which plasma generation source (or system) **416** extends. Fluid inlets **414** and **418** extend into chamber interior **401** through sidewalls (or a base) **405** of the chamber **400**. Fluid inlet **414** is in fluid communication with fluid source **420**, while fluid inlet **418** is in fluid communication with fluid source **422**. In other embodiment, fluid inlets **414** and **418** may be in fluid communication with the same fluid source. It should be appreciated that fluid inlets **414** and **418** may extend around a surface of the substrate **406** so that a perimeter of substrate **406** is encompassed by fluid inlets **414** and **418**. In some embodiments, fluid inlets **414** and **418** are configured as ring portions surrounding substrate **406**

[0050] In some embodiments, fluid inlets **414** and **418** are movable to vertically translate along with the substrate support **404** so that each fluid inlet remains proximate to an edge of substrate **406**. For example, the ring portions may be coupled to an appropriate drive such as a worm gear, linear drive, etc., so that the fluid inlets **414** and **418** track the movement of the substrate and substrate support.

[0051] The plasma generation source **416** is operable to provide a plasma activated species. "Plasma activated species" refers to reactive atomic and molecular radicals converted from the precursor gas through interaction with the plasma. The plasma also consists of non-charged species (e.g., radicals) and charged species (e.g., ions and electrons). The plasma activated species provided by plasma generation source **416** may have a non-reactive outer portion **440** surrounding a reactive inner portion **442**. Plasma generation source **416** may be a commercially available inductively coupled radio frequency (RF) plasma generation source. Plasma generation source (or system) **416** may include means for generating multiple types of plasma simultaneously.

[0052] Plasma activated species **440**, **442** exit plasma generation source **416** through showerhead body **436** and into showerhead **426**. Showerhead **426** diffuses the flow of plasma activated species through a number of injection ports into multiple paths **444**. Some plasma processes do not use showerheads. Showerhead body **436** and showerhead **426**, as illustrated, are suspended in chamber interior **401** above substrate **406**. Showerhead **426** and showerhead body **436** may be vertically translatable (i.e., movable) within chamber interior **401** by means of showerhead translator **434**. Showerhead translator **434** may include any appropriate drive such as a worm gear, linear drive, etc., and may be operable to translate showerhead **426** dynamically as processing continues.

[0053] An additional fluid source **428** may be coupled to (i.e., in fluid communication with) showerhead **426**. Fluid source **428** may provide, for example, an inert gas to the showerhead during processing. In some embodiments, the showerhead **426** is grounded. However, in other embodiments, a power supply (and controller) **430** may also be provided to control and modulate a charge on the showerhead **426** and/or control showerhead translator **434**. Alternatively, showerhead translator **434** may be controlled by controller **432** and powered by power supply **424**, or any other suitable source of power and control may be used.

[0054] To remove excess precursors, buffer gases, waste products, and other fluids from chamber interior **401**, one or more vacuum pumps **448** may be in fluid communication with chamber interior **401** via exhaust port **438**. Exhaust port **438** may be located on any convenient or effective area of chamber **400**. There may be multiple exhaust ports. In some embodiments, showerhead **426** or showerhead body **436** may have its own exhaust port, for example to exhaust fluids introduced into showerhead **426** by fluid source **428**. An exhaust may be a multi-port ring near the substrate (not shown) as a counterpart to distribution ring **415**. The operation of multiple exhausts may be independently and/or programmably controllable.

[0055] The chamber **400** also includes a controller (or control sub-system) **432** in operable communication with the other components of the chamber **400**, such as fluid sources **420**, **422**, and **428**, power supply **424**, etc. (for drawing simplicity, not all connections are shown). The controller **432** may include a processor, memory such as random access memory (RAM), and a storage device such as a hard disk drive. The controller **432** is configured to control the operation of the chamber **400** to perform the methods and processes described herein.

[0056] The embodiments illustrated in FIG. **4** provide for independent control of a plasma and a feedstock of a film to be deposited. For example, the plasma activated species provided by the plasma generation source **416** pass through showerhead body **436** and showerhead **426** into chamber interior **401**, while the film feedstock may be delivered through the bottom of the chamber to distribution ring **415** above or proximate to the substrate surface. In some embodiments, the distribution ring **415** is coupled to the substrate

support **404** so that the ring vertically translates with the substrate support. It should be appreciated that the feedstock interacts with the plasma proximate to a surface of substrate **406** so that site-isolated processing may be performed on different regions of substrate **406**. It should be further appreciated that the chamber **400** may be a vapor deposition chamber that includes chemical vapor deposition chambers and atomic layer deposition chambers.

[0057] In some embodiments, a plasma provided by plasma generation source **416** may include a plasma based on hydrogen, nitrogen, argon, oxygen, ammonia, nitrogen trifluoride, helium, or a combination thereof and may be referred to as a first precursor. A film feedstock provided through fluid inlets **414** and **418** may be any suitable feedstock for the desired deposition layer and may be referred to as a second precursor. In some embodiments, the first precursor carries the plasma activated species and activates the second precursor proximate to the substrate surface at a specific site or region. However, in some embodiments, only a first precursor is utilized without a second precursor.

[0058] In some embodiments, chamber **400** is used to perform a surface treatment on the substrate **406**. Some examples of surface treatments may preferentially expose substrate **406**, or a site-isolated region of substrate **406**, to a selected subset of plasma-activated species **440**, **442** generated by plasma generation source **416**. Some examples of surface treatments may include dynamically adjusting the concentration of one or more plasma-activated species at substrate **406** while the treatment process continues.

[0059] To reach substrate **406**, the plasma-activated species must pass through showerhead body **436**, showerhead **426**, and the part of chamber interior **401** between showerhead **426** and the surface of substrate **406**. Each plasma-activated species has an average expected lifetime after activation and an expected travel time from plasma generation source **416** to substrate **406**. In some cases, a species will only react with a substrate if the substrate provides an available reactive site (e.g., a dangling bond). The probability of impacts or reactions of an unwanted plasma-activated species on the substrate may therefore be reduced (the unwanted species can be "unselected") by decreasing the species' expected lifetime, increasing the expected travel time, and sometimes by removing reactive sites from the substrate (e.g., passivating the dangling bonds).

[0060] Factors influencing the lifetime of a species include intrinsic factors (e.g., reactivity of the particular species) and external factors (e.g., opportunities to be converted to another state by collision or reaction). Factors influencing the expected travel time from the plasma generation source to the substrate include travel velocity (which can be affected by pressure and temperature) and travel distance (which can be affected by hardware dimensions and mean free path between collisions). The probability of some species' reaching the substrate may be influenced by adjusting the composition, flow rate, pressure, or temperature of the plasma at the plasma generating source.

[0061] Adding a gas to the chamber increases the probability of collisions and, if the gas is reactive rather than being an inert "buffer" gas, conversion of some plasma-activated species to other species. The amount of gas added affects the travel time, and the type of gas added can affect the lifetime of a species with which the gas reacts. Changing the amount or type of added gas in the chamber during the plasma treatment is one way to dynamically change the selection of plasma activated species that reach the substrate. Adjusting the flow rate, pressure, or temperature of the added buffer or reactive gas can also affect the selection of species.

[0062] In the process chamber shown in FIG. **4**, gases can be introduced through a variety of inlets with potentially different effects. For example, a gas intended to bond with an unselected species or induce relaxation of the unselected species to a lower-energy state through collision may perform efficiently when introduced into the relatively confined space of showerhead body **436** or showerhead **426**, e.g., through fluid source **428**. A gas intended to mitigate certain types of reactive sites on the substrate may perform more efficiently if introduced through distribution ring **415** close to substrate **406**, or through fluid source **420** or fluid source **422**. Exhaust port(s) **438** may be located, and their timing operated, to confine a specific gas to a specific part of chamber **400** (e.g., showerhead **426** or the vicinity of substrate **406**) by exhausting it before it substantially diffuses throughout chamber interior **401**.

[0063] For example, a common plasma process in semiconductor manufacture converts an element (e.g., Ti) or an alloy or compound (e.g. TaSi) to its nitride (e.g., TiN, TaSiN) by exposure to plasma-activated N* radicals. N* radicals are preferred for some applications because higher-energy species such as ions, although they can cause a faster conversion, may unacceptably damage some surfaces (e.g., those of ALD films a few nanometers thick) by the amount of energy they dissipate. An extra process (e.g., passivation or planarization) is then required to repair the damaged surface before the next stage of fabrication can proceed.

[0064] Experiments indicated that introducing an inert gas (e.g., Ar, He in the plasma generation source) increases the concentration of N* radicals at the substrate and thereby increases the rate of nitride formation. Even in an excited state, Ar or He in a plasma generating source will not react with nitrogen; instead it transfers its collision energy to form more N* radicals. By contrast, introducing a reactive gas (e.g., H_2, O_2) has the opposite effect, quenching N* radicals and reducing their relative concentration. In addition, the presence of any extra gas may select longer-lived species by increasing collisions and shortening the mean free path. The increased collisions increase the mean travel time from the plasma generating source to the substrate. As mean travel time increases, more of the shorter-lived species decay, relax, react, or otherwise reach their end-of-life before reaching the substrate. In effect, the collision-increased travel time acts as a filter that prevents the shorter-lived species from reaching the substrate and still permits the longer-lived species to reach the substrate, such that the longer-lived species are selected and the shorter-lived species are unselected. The presence of a reactive gas that preferentially quenches or otherwise reacts with unselected species may change the states of those species before they reach the substrate.

[0065] FIGS. **5**A and **5**B are flowcharts of example processes for controlling a concentration of N* radicals reacting with a surface of a substrate. FIG. **5**A shows a process that includes real-time monitoring. This could be a nitride formation process or any type of N* treatment with results that can be monitored while the process is ongoing. The process is started **500** and the surface to be treated is exposed **501** to the nitrogen plasma. In some embodiments, an inert "buffer" gas may be added to the chamber to raise the pressure or increase the mean travel time by causing more collisions, thus preventing short-lived species from reaching the substrate. The

progress of the process (e.g., the conversion to nitride) is monitored **502**. Examples of monitoring **502** include, without limitation, measuring the sheet resistance of a film on the substrate that is affected by the treatment, or monitoring nitrogen peaks in a spectral trace (e.g., Fourier transform infrared (FTIR)). The sheet resistance may be measured, for example, by a diameter scan with a 4-point probe and a comparison with a previous scan. Taking a baseline scan before any treatment removes uncertainties based on unit-to-unit variations in film parameters such as thickness.

[0066] Until the process is complete, an indication of the concentration of N* radicals near the surface under treatment is monitored **503** and compared with a desired range. Examples of monitoring **503** the N* concentration include, without limitation, measuring a rate of change in the sheet resistance of the surface film (in processes where N* radicals are known to be the dominant cause of the change) or monitoring an emission spectrum of the plasma (which indicates how many N* radicals are being generated). An optical emission spectroscopy (OES) peak at 674.6 nm is widely used as an indicator of N*. This peak was observed to increase in intensity when inert gas (e.g., Ar) was added and decrease when H_2 or O_2 were added.

[0067] If the monitoring indicates **505** that not enough N* radicals are reaching or reacting with the surface, then inert gas (e.g., Ar or He) is added or reactive gas (e.g., H_2 or O_2), if present, is exhausted **506**. If, instead, the monitoring indicates **507** that too many N* radicals are reaching or reacting with the surface, then reactive gas (e.g., H_2 or O_2), is added or inert gas (e.g., Ar or He), if present, is exhausted **508**. The N* radicals are treated as a selected species if the reaction is proceeding more slowly than a desired rate; the same N* radicals are treated as an unselected species if the reaction is proceeding more quickly than the desired rate. After adjusting the gas mixture, or if the concentration of N* radicals appears to be within the desired range, the exposure and monitoring continue **509→501**, **502** until the process is complete **503**, at which point the process is ended **510**. Because the N* radicals do not dissipate much extra energy beyond what is necessary to create the nitride, an extra repair step is not needed after the nitridation.

[0068] HPC can be helpful in optimizing this type of process. For example, some of these effects have thresholds. Adding Ar to the N_2 buffer gas during the nitride conversion of Ti at 0.16 Torr only begins to increase the change in sheet resistance of the Ti film (evidence of more Ti being converted to less-conductive TiN) when the concentration ratio of N_2/Ar exceeded about 30%.

[0069] FIG. **5**B illustrates a process for HPC optimization of an N* treatment process without real-time monitoring. A number of site-isolated regions (SIRs) are defined on the substrate to be processed under varying conditions. Some SIRs may be set aside to be identically processed to measure the repeatability or spatial uniformity of a "control" process. The conditions to be varied may include substrate temperature, chamber pressure, plasma precursor composition, plasma power (e.g., DC or RF power), exposure time, flow rates of process gases, and the relative concentrations of inert and reactive gases. After the process is started **520**, a set of these process conditions is selected **521** for each SIR, and the SIR is exposed to the plasma under that selected set of conditions. The area of the substrate outside the currently processed SIR may be, for example, shielded by a mask so that only the SIR is exposed. If there are still SIRs unprocessed **523**, the next SIR is prepared **524** for exposure under the next set of conditions (e.g., the chamber is purged and (a) the mask is moved or (b) the substrate is moved under a stationary mask).

[0070] When all the SIRs are processed **523**, they are characterized **525** (e.g., their sheet resistance, I-V curves, C-V curves, reflectance spectra, or other properties are measured). By comparing the measurements of the SIRs in view of a desired property (e.g., high or low conductivity or nitrogen content), the set of conditions that produced the best results can be identified **530**.

[0071] The dynamics of fluids (including gases) in the chamber also affect the relative concentrations of species reaching the substrate. These effects may be largely independent of composition and can be manipulated separately via the length and "conductance" (which herein refers to fluid/gas conductance, not electrical conductance) of the passages from the remote plasma to the substrate. Referring back to FIG. **4**, the passages include remote plasma generation source **416**, showerhead body **436**, showerhead **426**, and chamber interior **401**, as well as any intervening conduits present in various chamber embodiments but not illustrated in FIG. **4**.

[0072] FIGS. **6**A-**6**C illustrate embodiments of showerheads and their injection ports. FIG. **6**A is a top perspective view of a showerhead embodiment. To view the internal details, the showerhead body is not shown. Showerhead **600** may be formed from any known suitably inert materials, such as stainless steel, aluminum, anodized aluminum, nickel, ceramics and the like. Showerhead **600** is substantially circular. Its outer diameter may be about 200 or 300 mm, or up to 600 mm or even larger, depending on the substrate size. Other sizes or shapes may be used; for example, to match differently sized substrates. A plurality of injection ports (or openings) **602** extend through a perforated wall **604** of the showerhead **600**. Showerheads diffuse the plasma-generated species entering the main process chamber interior (**401** in FIG. **4**).

[0073] A fluid separation mechanism **606** extends upwards from the perforated wall **604** and includes several substantially linear dividers to divide the perforated wall **604** into four quadrants **608**, **610**, **612**, and **614**. In some embodiments, quadrants **608**, **610**, **612**, and **614** correspond to similarly shaped, site-isolated regions on the substrate (**406** in FIG. **4**). In some embodiments, perforated wall **604** may be divided into a different number of sections (e.g. 2, 3, 6, 8, or any suitable number). The height of fluid separation mechanism **606** above a top surface of perforated wall **604** depends on showerhead design parameters; in some embodiments, fluid separation mechanism **606** provides sufficient separation to minimize or prevent diffusion of fluids between adjacent quadrants **608**, **610**, **612**, and **614**, thus facilitating combinatorial processing of corresponding regions on the substrate. A fluid trap ring **616**, extending upwards from a periphery of perforated wall **604**, may assist in containing fluid within showerhead **600**.

[0074] FIG. **6**B is a bottom perspective view of a showerhead with a different hole pattern. Injection ports **602** need not be arranged in a uniform rectilinear pattern as in FIG. **6**A. Here, injection ports **602** are arranged annularly and only cover part of the available area of perforated wall **604**. Injection ports in some embodiments may be spaced regularly or irregularly and may have different diameters or different shapes on the same showerhead. In some embodiments, the arrangement of injection ports **602** may differ between the

different quadrants **608-614**. In some embodiments, the showerhead is not divided into quadrants or other sections.

[0075] FIG. **6**C is a magnified partial sectional view of an injection port through the section A-A in FIG. **6**C. The direction of flow through the showerhead is indicated by arrows **622**. Parameters that can be varied in injection port **602** include its length L, its bore angle α, its input width W_i, its output width W_o, and characteristics of its internal wall **603** such as taper, curvature, or texture.

[0076] Likewise, each space that the plasma activated species flow through has conductance properties determined by parameters such as the dimensions and shape of the space. Some of these may be dynamically variable. For instance, showerhead **426** in FIG. **4** can be moved by showerhead translator **434** to lengthen the part of the path preceding the showerhead and shorten the part of the path following the showerhead, or vice versa. Apertures affecting conductance can be inserted in or removed from parts of the path, or their diameters can be changed similarly to the variable aperture in an iris diaphragm.

[0077] Changes in conductance along the path traversed by the plasma activated species affects the relative concentrations of species reaching the substrate by affecting (1) the pressure at the plasma generation source, which can affect the relative concentrations of species being generated, and (2) the travel time for the species to reach the substrate, which begins to exclude species with expected lifetimes shorter than the travel time. Thus, the relative deposition rate of selected species and unselected species from a plasma can be manipulated by changing the dimensions and geometries of the showerhead holes and the remote-plasma passage into the substrate chamber. Sheet resistance of an underlying Ti film and/or optical emission spectroscopy of the plasma source can be used to evaluate the results of these changes.

[0078] For example, in a conversion of a metal, semiconductor, or mixed substance to its nitride by N* radicals, the nitride conversion proceeded at a lower rate when the injection ports of the showerhead had a smaller diameter. Other processes besides nitridation (e.g. doping) can also use these approaches to select or include certain plasma activated species to react with the substrate.

[0079] One type of plasma treatment improved by selection of species is the removal of native oxides from substances such as Ge and III-V materials. FIGS. **7**A-**7**C conceptually illustrate the formation and removal of a native oxide. Many materials, such as semiconductor material **701** (whether in bulk or thin-film form) spontaneously react with oxygen and/or water vapor in the ambient atmosphere to form a native oxide **702**. Unfortunately, even a few A of native oxide **702** can compromise the performance of a gate stack or source-drain contact fabricated on or through it. Therefore, before fabrication can proceed, native oxide **702** must be removed somehow to expose a pure material surface **703**, and generally surface **703** also needs to be passivated to sequester any surface defects.

[0080] Germanium (Ge) and III-V materials (e.g. GaN, GaAs, InP) form native oxides, but are unacceptably damaged by the ion bombardment techniques used for removing similar native oxides from other materials. Silicon (Si), for example, can tolerate oxide removal by plasma treatment with higher-energy species because its electron energy levels are further below the Fermi level and its oxides are comparatively stable and self-limiting. By contrast, Ge and the III-V materials have electron energy levels that are closer to the Fermi level, and therefore they behave more like metals; their oxides are unstable and do not self-limit, and bombardment with high-energy species is likely to leave dangling bonds and other defects. O* and H* radicals, on the other hand, are observed to remove the oxides and passivate the surface without damage.

[0081] FIG. **8** is a flowchart of an example process for native-oxide removal using O* and H* radicals as the selected species. A pre-conditioning step **801** removes trapped water from the showerhead and other chamber hardware such as the process kit. Typically, temperature ranges from about 120-400 C, chamber pressure ranges from about 0.5-1 torr, flow rates are about 500 sccm, and durations are 10-30 min for the pre-conditioning step, which may optionally include an Ar or N_2 buffer gas. A residual gas analyzer (RGA) may optionally be used 811 to verify removal of the trapped water. After pre-conditioning **801**, an etch rate of the oxide is monitored (**802**). Monitoring techniques may include tracking a known oxide FTIR peak near the surface being treated; for example, GeO2 has characteristic infrared absorption peaks at 560 and 870 cm^{-1}.

[0082] Monitoring continues or is repeated while subjecting the surface to alternating periods of Treatment A (**803**) and Treatment B (**804**). Treatment A may be exposure of the surface to either H* or O* radicals. Treatment B is exposure of the surface to the other type of radical; that is, if Treatment A uses H*, Treatment B uses O* and vice versa. For Treatments A and B, the O* and H* temperatures may range from about 150-250 C, the chamber temperature may range from about 80-120 C, and the chamber base pressure may be about 2e-6 Torr. When the monitoring results indicate that the oxide is removed (e.g., the oxide peak is no longer detectable or falls below a predetermined threshold representing an acceptable trace level), the process is ended. The surface is now free of oxide within the pre-defined tolerance, and also passivated and ready for the next fabrication step (e.g., formation of a source, drain, or gate). This method is suitable for general cleaning of impurities from Ge, III-V materials, and other materials that tend to sustain damage from ion bombardment.

[0083] The process may be optimized for a substrate type or pre-defined tolerance by HPC. The results of the HPC samples may be compared by observing the growth of ALD films on the cleaned surface. Alternatively, the results may be compared by depositing a metal or other conductive film on the cleaned surface ("capping") and either (1) forming a diode, measuring its I-V curve, and comparing it with the I-V curve of a diode formed from a known clean sample, or (2) measuring the line resistance of the conductive material.

[0084] Although the foregoing examples have been described in some detail to aid understanding, the invention is not limited to the details in the description and drawings. The examples are illustrative, not restrictive. There are many alternative ways of implementing the invention. Various aspects or components of the described embodiments may be used singly or in any combination. The scope is limited only by the claims, which encompass numerous alternatives, modifications, and equivalents.

What is claimed is:

1. A method of plasma-treating a surface of a substrate, the method comprising:

positioning the substrate in a process chamber;

creating a plurality of plasma activated species;

selecting a species from the plurality of the plasma activated species while leaving another species unselected; and

preferentially exposing the surface to the selected species by modifying at least one of a relative concentration of the selected species and the unselected species, an expected lifetime of the unselected species, or an expected travel time from a plasma generating source to the surface.

2. The method of claim **1**, wherein the selected species comprises a radical.

3. The method of claim **3**, wherein the selected species comprises a nitrogen radical.

4. The method of claim **3**, wherein the selected species comprises an oxygen radical or a hydrogen radical.

5. The method of claim **1**, wherein the selected species is selected for low energy dissipation.

6. The method of claim **5**, wherein the surface comprises a layer formed by atomic layer deposition.

7. The method of claim **6**, wherein the layer is less than about 10 nm thick.

8. The method of claim **5**, wherein the surface comprises germanium, germanium oxide, a III-V material, or a III-V material oxide.

9. The method of claim **1**, wherein the relative concentration of the selected species and the unselected species is modified at the plasma generating source.

10. The method of claim **9**, wherein modifying the relative concentration comprises introducing an inert gas.

11. The method of claim **10**, wherein the inert gas comprises argon or helium.

12. The method of claim **9**, wherein modifying the relative concentration comprises introducing a reactive gas.

13. The method of claim **12**, wherein the reactive gas comprises oxygen or hydrogen.

14. The method of claim **12**, wherein the reactive gas preferentially quenches the unselected species compared to the selected species.

15. The method of claim **1**, wherein modifying the expected lifetime or the expected travel time comprises adjusting a composition, flow rate, pressure, or temperature of a buffer gas.

16. The method of claim **1**, wherein modifying the expected lifetime or the expected travel time comprises adjusting a composition, flow rate, pressure, or temperature of a plasma at the plasma generating source.

17. The method of claim **1**, wherein modifying the expected lifetime or the expected travel time comprises changing a gas conductance of the plasma generation source, a showerhead body, a showerhead, or a chamber interior.

18. The method of claim **1**, wherein modifying the expected lifetime or the expected travel time comprises changing a gas conductance of a showerhead by changing the size, position, or geometry of an injection port in the showerhead.

19. The method of claim **1**, further comprising measuring an effect of the plasma activated species on the surface by a method comprising one of:

a sheet resistance of a film over or under the surface;

a line resistance of a feature on the surface;

an I-V curve of a diode created by depositing a conductive film on the surface;

a C-V curve of a capacitor structure created on the surface;

a characteristic of an atomic-layer-deposition film on the surface

a Fourier transform infrared spectrum taken near the surface; or

an optical emission spectrum of the plasma generating source.

20. A method of cleaning a surface of a substrate, comprising:

positioning the substrate in a process chamber;

pre-conditioning by removing trapped water from the substrate and from the process chamber; and

exposing the surface to an alternating sequence of O* and H* radicals from a plasma generating source until the surface is clean;

wherein higher-energy species than the O* and H* radicals are generated by the plasma generating source but are prevented from reaching the surface.

* * * * *